FROM AI EXPOSURE TO LABOR FUTURES: HOW INSTITUTIONAL ARRANGEMENTS SHAPE THE FUTURE OF WORK

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ABSTRACT

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by

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Public discussion of artificial intelligence and work often treats technological capability and occupational exposure as direct indicators of future labor outcomes. This capstone challenges that assumption by asking: How do institutional arrangements shape divergent labor futures under conditions of AI exposure? It argues that the effects of technological change on work are institutionally mediated rather than technologically predetermined.

The project combines research on AI and labor, institutional theory, comparative policy analysis, occupational and task data, and scenario-based reasoning. AI exposure measures are treated as a technological starting point rather than as forecasts of displacement. The analysis instead focuses on the processes that intervene between technological capability and labor-market outcomes: institutional arrangements, organizational responses, and choices about how human and AI capabilities are combined in the organization of work. Comparative analysis of the United States, Singapore, and Denmark illustrates how differences in labor-market institutions, workforce-development systems, governance, social protection, and employer incentives can shape different responses to technological change.

These ideas are translated into the Institutional Scenario Explorer for AI & Labor Futures, an interactive web-based artifact that allows users to hold occupational AI exposure constant while changing institutional and organizational conditions and examining how resulting pressures on labor outcomes move in different directions. The explorer is not a predictive simulator. Its scenarios are evidence-informed qualitative syntheses designed to make assumptions, trade-offs, and causal pathways visible rather than to forecast a single future.

The project demonstrates the limits of interpreting AI exposure as destiny. While technological exposure can be quantified with considerable precision, the social and organizational processes that determine what follows are more contingent and difficult to measure. By making those mediating choices explicit and interactive, the project offers a framework for thinking about AI and work as a problem of institutional design, organizational decision-making, and distribution, rather than technology alone.

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I would also like to acknowledge the intellectual influence of Daron Acemoglu and Simon Johnson's *Power and Progress*. The book fundamentally changed the way I think about technology and its consequences for society, the economy, and work. It connected questions that I had previously encountered separately and showed me how closely technological change is tied to institutions, power, and choices about who benefits. That perspective encouraged me to zoom out, and resist being pulled along by the dizzying speed and hype of contemporary AI discourse. In many ways, it provided the intellectual starting point for this project.

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INTRODUCTION

Artificial intelligence has become a central frame through which the future of work is discussed. Much of this debate moves between expectations of far-reaching transformation and skepticism that AI will produce more than limited efficiency gains and new inequalities. Jungherr (2026) argues that both positions can obscure the mechanisms through which technological change actually produces social effects. Rather than treating these effects as automatic, his framework describes AI-driven change as contingent, stepwise, and mediated by processes including organizational adoption, shifts in relative power, governance, and existing structural constraints. This distinction is especially important in discussions of AI and labor, where technological capability and eventual labor-market outcomes are often treated as if they were the same thing.

The ability of an AI system to perform or assist with a task does not, by itself, determine how that capability will be used in the workplace. Organizations still make decisions about adoption, task allocation, work reorganization, training, and redeployment, and those decisions take place within institutional environments that shape available options and incentives. Workforce-development research provides one example: employer and worker responses to technological change are shaped by labor-market structures, training systems, public funding, and the distribution of bargaining power (Altman & Schrag, 2025).

This white paper develops that argument through the Institutional Scenario Explorer for AI & Labor Futures, an interactive project built around the research question: How do institutional arrangements shape divergent labor futures under conditions of AI exposure? The project treats occupation-specific AI exposure as a baseline rather than a forecast. It then examines how different configurations of labor institutions, workforce-development systems, governance, social protection, and organizational choices can alter the pathway from technological capability to work. Rather than predicting a single future of AI and labor, the Scenario Explorer is designed to make these mediating choices visible and to explore why similar technological conditions can plausibly lead to different labor outcomes.

The project's contribution is therefore less about producing another estimate of how many jobs may be affected by AI and more about making the intervening mechanisms visible. It separates five analytically distinct elements: occupational and task exposure, the institutional

environment, organizational response, human–AI work design, and labor-market outcomes. Exposure identifies where AI capabilities may intersect with work; institutional arrangements shape the conditions surrounding that exposure; organizations decide how to respond within those conditions; work-design choices determine how tasks are reorganized between people and AI; and labor outcomes emerge from the interaction of these processes. The Scenario Explorer operationalizes this framework as an interactive analytical tool rather than a predictive model.

The remainder of the paper develops this framework in stages. The literature review examines task-based technological change, organizational adoption, human–AI work design, institutional mediation, and technological expectations. The conceptual framework translates these literatures into the analytical structure of the Scenario Explorer. The methodology section describes the occupational and exposure data, evidence base, scenario construction, work-design logic, and outcome synthesis. The paper then situates the institutional dimensions in comparative context, explains how the conceptual model was translated into the interactive artifact, and discusses the project's implications.

1. LITERATURE REVIEW: FROM TECHNOLOGICAL CAPABILITY TO LABOR OUTCOMES

Research on technology and work suggests that technological capability alone is insufficient to explain labor outcomes. The literature reviewed here identifies several intervening mechanisms, including task-level substitution and complementarity, organizational adoption, human–AI work design, institutional conditions, and expectations about technological futures.

1.1 Tasks, Automation, and Labor Demand

A foundational insight in research on automation is that jobs should not be treated as indivisible units. Autor (2015) argues that occupations consist of multiple tasks and that technology can substitute for human labor in some tasks while complementing it in others. Automation may make particular activities cheaper or faster while increasing the value of tasks that remain with workers. Its labor-market effects consequently depend not only on technical substitution but also on complementarities between tasks, changes in demand, and labor-

supply responses. Autor also cautions against explaining occupational change through technology alone, pointing to the simultaneous influence of macroeconomic conditions and labor-market institutions. The important implication for this project is that the technological susceptibility of some tasks within an occupation does not, on its own, establish the fate of the occupation or the workers within it (Autor, 2015).

For this project, that distinction also has an important implication for agency: explaining occupational change through technology alone makes workers and institutions appear more passive than they are. It obscures the choices about adoption, work design, training, and protection through which technological change is translated into labor outcomes.

Acemoglu and Restrepo (2019) develop this task-based approach more formally. Their framework treats production as a collection of tasks allocated between labor and capital. Automation occurs when new technologies allow capital to perform tasks previously carried out by workers, changing what the authors call the task content of production. This produces a displacement effect, shifting the allocation of tasks away from labor. At the same time, automation can generate a productivity effect by lowering production costs and increasing output, potentially increasing demand for labor in tasks that remain with workers. The net effect on labor demand therefore depends on the relative strength of these mechanisms rather than following automatically from the introduction of an automation technology (Acemoglu & Restrepo, 2019).

The displacement–productivity distinction may not capture every way that lower-cost task completion changes work. AI can also make it feasible to perform tasks that previously remained undone because they fell too far down an organization's priority list. This may partly appear as a productivity gain, but it also expands the scope of work by making more tasks worth undertaking in the first place.

A further mechanism is reinstatement: technological change can create new tasks in which labor has a comparative advantage, expanding the range of activities performed by workers. Acemoglu and Restrepo's empirical decomposition suggests that displacement and reinstatement were approximately offsetting in the United States between 1947 and 1987, whereas 1987–2017 was characterized by stronger displacement and weaker reinstatement, alongside slower productivity growth. These estimates are produced through the authors' decomposition framework rather than directly observed causal effects, but they illustrate why

the labor consequences of technological change depend on the type and direction of innovation rather than automation alone (Acemoglu & Restrepo, 2019).

For this project, the task-based literature establishes the first distinction between what a technology may be capable of doing and what eventually happens to an occupation.

1.2 From Technological Capability to Adoption and Organizational Change

Jungherr (2026) addresses the next step in a broader theory of AI-induced social change. Rather than treating technological innovation as sufficient to produce social effects, he conceptualizes AI-driven change as contingent, stepwise, and mediated. His framework traces a sequence from technological innovation through adoption by organizations and individuals, uneven gains in relative advantage, shifts in power, governance responses, and material, institutional, and societal constraints. Each stage can redirect or interrupt the pathway from technological possibility to broader social change (Jungherr, 2026).

A particularly relevant distinction is between the availability of a technological capability and its adoption. The existence of an innovation does not ensure that actors will identify a useful application or integrate it into existing practices. Organizations vary in their resources, willingness to experiment, and capacity to adjust structures and processes. Adoption can therefore be uneven even where similar technological capabilities are available.

For the analysis of AI and work, this means that technical capability alone does not specify whether an employer will replace a task, use AI to support an existing worker, reorganize a workflow, or decline to deploy the technology. Jungherr also emphasizes that these decisions occur within regulatory, institutional, material, and social constraints. His framework therefore addresses questions that task-based models leave open: why organizations adopt the same technological capabilities differently, and how organizational and institutional conditions shape their effects.

1.3 Human–AI Work Design and Skill Formation

Singh, Ghosh, and Dai (2026) examine how firms allocate work between AI systems and workers. Their theoretical model identifies a dynamic trade-off: relying more heavily on AI may improve short-term output when the technology performs better, but reduced worker engagement can also erode human skill. Keeping workers meaningfully involved can therefore

preserve or develop capabilities that may remain valuable when AI is imperfect or requires human fallback (Singh et al., 2026).

This makes human–AI work design a human-capital investment problem, rather than only an immediate efficiency decision. The model distinguishes between two dimensions of AI improvement: capability, or how well the system performs when it works, and reliability, or how consistently it works without failing or requiring human intervention. The distinction matters because the two can create different incentives for keeping workers engaged. A highly capable AI may outperform a worker on a task, but if it remains imperfect, workers may still need to retain the skills required to step in when it fails. As reliability improves and those failures become less frequent, the immediate need for this human fallback may decline. Singh et al. therefore show that treating AI progress as a single measure can obscure different, and sometimes opposing, effects on how work is divided between people and AI.

The model also considers worker mobility, meaning that workers can leave one employer for another and carry the skills they have developed with them. This complicates the firm's decision to invest in worker engagement: an employer may bear the cost of maintaining or developing a worker's skills without capturing all of the future benefit if that worker leaves. At the same time, jobs that offer stronger opportunities to develop valuable skills may themselves become more attractive to workers. Singh et al. therefore show that human–AI work design can influence both the skills a firm has available in the future and the value workers place on different job opportunities (Singh et al., 2026).

Because the paper is a theoretical model, its contribution here is conceptual rather than evidence that firms currently behave in this way. It identifies a mechanism through which organizational choices about human–AI work design can shape later worker capabilities and outcomes.

1.4 Institutions and Labor-Market Adjustment

Institutional approaches shift attention from individual or organizational choices to the rules and systems that shape the conditions under which those choices are made. North (1990, p. 3) defines institutions as the "rules of the game" that structure incentives in political, social, and economic interaction, emphasizing both formal rules and informal constraints. Hodgson

(2006) similarly defines institutions as established social rule systems that structure interaction, while stressing that institutions can both constrain and enable action.

Altman and Schrag (2025) provide a concrete example of how such conditions shape labor-market adjustment. Their analysis of U.S. workforce development shows that training and skill formation are organized across employers, government programs, educational institutions, workforce organizations, and other intermediaries. Employer incentives to invest in training depend partly on worker turnover and the likelihood that firms can capture returns from that investment, while apprenticeship systems, credentials, public funding, and labor-market rules affect how skills are developed and whether they are portable across employers.

Labor-market adjustment is therefore more than an individual response to technological change. Access to training, incentives for employer investment, and workers' ability to move into new forms of work are shaped by existing institutional conditions. Taken together, this institutional perspective suggests that adaptation capacity is not simply a property of individual workers or firms, but of the systems that shape training, mobility, and adjustment.

1.5 Expectations, Hype, and Technological Futures

Technological change is also shaped by expectations formed before its eventual effects can be known. Borup, Brown, Konrad, and Van Lente (2006) argue that expectations in science and technology are not merely forecasts. They can be generative, helping to guide activity, attract interest and investment, legitimate action, and coordinate actors around particular technological futures. Expectations therefore become part of the innovation process rather than simply predictions against which later outcomes can be evaluated (Borup et al., 2006).

The sociology of expectations also draws attention to cycles of promise, hype, and disappointment. Borup et al. show how expectations change over time and across actors, and how dominant visions can shape which technological pathways receive attention and resources. This perspective is particularly relevant where technological futures remain uncertain but consequential decisions are already being made.

Galanos et al. (2026) explicitly position Critical Hype Studies (CHS) as building on the sociology of expectations and related approaches to technological futures. Their position paper argues that hype should not be understood simply as exaggeration or inaccurate prediction,

but as a sociotechnical phenomenon intertwined with funding, legitimacy, and power. Hype can mobilize attention and resources while also narrowing which futures are treated as possible or fundable (Galanos et al., 2026).

This perspective is relevant to AI and labor because claims about automation, productivity, and worker displacement can influence investment, organizational strategy, and public policy before their eventual labor-market effects are known. Galanos et al. do not empirically establish that particular AI forecasts produce particular employment outcomes; their contribution here is conceptual. Together with Borup et al., their work provides a basis for treating narratives about technological futures as part of the process being studied rather than as neutral descriptions of what technology will inevitably produce.

1.6 Synthesis and Research Gap

Taken together, these literatures identify distinct mechanisms between technological capability and labor outcomes: task-level substitution and complementarity, organizational adoption, human–AI work design, institutional conditions, and expectations about technological futures. These mechanisms operate at different analytical levels and do not form a single causal model. Their differences are useful because they expose processes that can otherwise be compressed into the claim that an occupation is simply "exposed" to AI.

The Institutional Scenario Explorer for AI & Labor Futures brings these mechanisms into a common analytical structure without attempting to combine them into a predictive model. It separates technological exposure, institutional conditions, organizational response, human–AI work design, and labor outcomes so that their relationships can be explored without treating them as interchangeable. More broadly, this structure also pushes back against a more algorithmic way of understanding complex social change, in which multiple conditions are reduced to inputs that produce a single outcome.

2. CONCEPTUAL FRAMEWORK: INSTITUTIONAL MEDIATION OF AI AND WORK

The conceptual framework translates the mechanisms identified in the literature review into the analytical structure of the Scenario Explorer. It separates AI exposure, institutional

conditions, organizational decisions, human–AI work design, and labor outcomes so that each can be examined without treating any single layer as determinative.

2.1 Institutions, Organizations, and Institutional Arrangements

Building on the institutional definitions established in the literature review, this project treats institutions as the rules and systems that structure interaction and organizations as actors operating within those broader conditions. This is an analytical distinction: as Hodgson (2006) notes, organizations contain their own internal rules, but they can also be treated as actors within wider institutional systems.

For the purposes of the Scenario Explorer, a labor union, government agency, university, or firm is therefore treated as an organization, while collective bargaining rules, employment protections, credentialing systems, or unemployment insurance are treated as institutions. The distinction allows employer choices to remain analytically separate from the broader conditions that shape the options and incentives employers face.

The project uses institutional arrangement to refer to the configuration of institutions, organizations, and relationships through which labor-market adjustment is structured. This is an operational definition developed for the Scenario Explorer rather than a definition taken directly from North or Hodgson. A workforce-development system, for example, may combine public funding, training providers, employers, credentialing arrangements, labor organizations, and worker protections. The relevant unit is not any one policy or organization, but how these elements operate together.

2.2 AI Exposure as the Baseline Condition

The Scenario Explorer treats AI exposure as a starting condition rather than an outcome. Exposure indicates where current AI capabilities may intersect with the tasks that make up an occupation. It does not indicate whether those tasks will actually be automated, how employers will reorganize work, or what will ultimately happen to employment, job quality, displacement pressure, or other labor-market outcomes. This distinction follows from the task-based literature reviewed above (Autor, 2015; Acemoglu & Restrepo, 2019).

The occupation selected by the user determines the technological and task context on which the rest of the scenario operates. The institutional environment and employer response

are varied separately, allowing the Explorer to examine different pathways under the same occupational baseline.

Exposure is therefore held conceptually constant as the user moves through a scenario. The specific data and methodology used to represent occupation-level AI exposure are discussed in Section 4.

2.3 Institutional Environment

The institutional environment represents the broader conditions within which employers and workers respond to AI exposure. In the Scenario Explorer, these conditions are represented through three broad institutional configurations: Industry-led adaptation, Publicly coordinated adaptation, and Worker-negotiated adaptation. The configurations vary across dimensions related to AI governance, worker protections, collective bargaining, workforce development, employer retraining incentives, active labor-market policy, and social protection. Together, these dimensions capture different combinations of rules, public capacities, and incentives that can shape how technological change is absorbed.

The scenarios are designed as institutional arrangements rather than individual policy interventions. They are intentionally not occupation-specific, and a particular configuration does not dictate how an employer will respond. Instead, it changes the protections, resources, incentives, and constraints surrounding organizational choices.

2.4 Organizational Response

The organizational response layer captures the choices employers make within a given institutional environment. Institutions can alter the conditions surrounding those choices, but organizations facing similar technological opportunities may still respond differently depending on their strategies, capacities, and willingness to reorganize around new technologies (Jungherr, 2026).

In the Explorer, organizational response is organized around three broad starting orientations: Automation-oriented, Hybrid work redesign, and Augmentation + redeployment. These orientations represent different combinations of three underlying organizational choices: deployment strategy, retraining and role transition, and worker participation in work redesign. They are starting configurations rather than fixed categories, allowing users to adjust the underlying choices independently.

The selected occupation determines the work context on which these choices operate. The occupation shapes the relevant task structure and exposure baseline, while organizational response represents a separate set of decisions about how AI is deployed, whether workers are retrained or redeployed, and how much they participate in redesigning work.

2.5 Human–AI Work Design

The human–AI work design layer focuses on how work is divided between workers and AI once an organization has chosen an overall approach to adoption. It captures which tasks remain primarily human, which are delegated to AI, and where work is shared or augmented.

Building on the human-capital mechanism discussed in the literature review, the Scenario Explorer also treats worker involvement in AI-assisted work as consequential for skill formation over time. Work design therefore captures not only whether tasks are automated or augmented, but whether workers remain meaningfully engaged in activities through which skills can be maintained or developed.

In the Explorer, this layer is represented through the Work Design Studio, which translates the employer's broader response into an occupation-specific pattern of task allocation across three modes: Human-led, Human + AI, and AI-led. Automation and augmentation are treated as choices about how work is organized rather than inevitable consequences of technological capability.

2.6 Labor-Market Outcomes

The final layer captures the labor-market consequences that emerge from the preceding stages of the framework. These include changes in employment, productivity, displacement pressure, skill formation, job quality, redeployment capacity, and the distribution of gains. In the Scenario Explorer, these outcomes are not treated as direct effects of AI exposure. They reflect the combined influence of the technological baseline, the institutional environment, employer decisions, and the resulting organization of human and AI work.

These outcomes are deliberately treated as separate dimensions rather than as a single measure of success or failure. The same pathway may improve productivity while increasing displacement, or support redeployment while distributing gains unevenly across workers. Making these tensions visible is important because labor-market adjustment involves trade-offs among outcomes that do not necessarily move together. The Explorer therefore does not

collapse results into a single favorable or unfavorable assessment but represents how different dimensions may move in different directions.

3. METHODOLOGY

The Scenario Explorer combines quantitative occupational data with coded academic and policy evidence in a structured scenario-based model. Occupational data establish the technological and task baseline, while the evidence layer informs the institutional, organizational, and outcome logic described below.

3.1 Occupational and AI-Exposure Data

The occupational data are drawn primarily from the Occupational Information Network (O*NET), the U.S. Department of Labor's standardized system for describing occupations and the tasks, skills, and work activities associated with them (National Center for O*NET Development, 2026). O*NET provides the common occupational structure used throughout the Explorer. Each occupation is identified by its O*NET-SOC code and linked to its description, task statements, task ratings, skills, work activities, and job-zone information. For the interactive display, tasks are ranked using O*NET measures of task type, importance, and frequency, with the highest-ranked tasks used to represent the occupation in the Work Design Studio. O*NET job-title data are also used to improve occupation search while retaining O*NET-supported occupational labels.

The app supplements these occupational profiles with national employment and wage information from the Bureau of Labor Statistics Occupational Employment and Wage Statistics (OEWS). These figures provide descriptive labor-market context for the selected occupation; they are not inputs into the scenario outcome rules.

AI exposure is represented using two existing occupation-level measures: the AI Occupational Exposure (AIOE) dataset developed by Felten, Raj, and Seamans (2021) and the OECD AI exposure measure, which is constructed from an AI Capability Gap index (Organisation for Economic Co-operation and Development [OECD], 2026a). In the current implementation, the AIOE score is converted into an occupational percentile, while the OECD measure uses its reversed normalized score so that higher values correspond to greater

alignment between AI capabilities and occupational requirements. Where both measures are available, the Explorer averages the two normalized values to create a combined exposure baseline. Scores below .33 are classified as low exposure, scores from .33 to below .66 as medium, and scores of .66 or above as high.

This combined measure is used as a technological baseline, not as an estimate of automation or job loss. It indicates relative exposure to AI capabilities at the occupational level while leaving subsequent adoption, task allocation, institutional conditions, and organizational choices to separate parts of the model.

The underlying data pipeline keeps original source files separate from cleaned and app-ready data. Raw occupational and exposure datasets are transformed through reproducible scripts into processed occupation profiles and exposure baselines used by the web application. This structure also separates quantitative baseline data from the literature-based evidence used in the scenario model.

3.2 Evidence Base and Coding

The Scenario Explorer organizes academic and policy research in a claim-level evidence matrix that informs the scenario logic. Rather than assigning an entire source to a particular scenario, individual claims are coded according to the mechanism they describe, the relevant outcome, its directional implication, geographic context, and the strength or verification status of the evidence.

This claim-level approach is intended to preserve traceability between the literature and the model. It also allows theoretical, empirical, and context-specific evidence to be distinguished rather than treated as equally generalizable. The coded claims inform the directional rules used to synthesize outcomes in the final stage of the Explorer.

3.3 Construction of Institutional Scenarios

The institutional layer is organized around three starting configurations: Industry-led adaptation, Publicly coordinated adaptation, and Worker-negotiated adaptation. Each represents a different combination of eleven dimensions: AI governance, algorithmic transparency, worker information rights, data protection, anti-discrimination rules, collective bargaining, workforce development, employer retraining incentives, active labor-market policy, social protection, and employment protection.

The configurations are analytical scenarios rather than representations of particular countries or predictions about institutional development. Users can modify individual dimensions, allowing the Explorer to examine alternative combinations of institutional conditions rather than treating the three starting configurations as fixed models.

3.4 Organizational Response and Work Design

The organizational layer begins with three starting configurations: Automation-oriented, Hybrid work redesign, and Augmentation + redeployment. Each represents a different combination of three adjustable organizational choices: deployment strategy, retraining and role transition, and worker participation in work redesign. Users can modify these dimensions independently, allowing the same institutional environment to support different organizational responses.

The Work Design Studio then translates the selected occupation and scenario into an illustrative allocation of occupation-specific tasks across three modes: Human-led, Human + AI, and AI-led. The tasks themselves come from the O*NET occupational profile described above. The initial placement is scenario-informed, but it is not intended as an estimate of the percentage of work that will be automated or augmented. Users can move tasks between the three modes to explore alternative ways of organizing the same occupation.

This interactive step keeps work design distinct from both technological exposure and the employer's broader response. Task allocation is therefore treated as part of the scenario design rather than inferred directly from an occupation's exposure score.

3.5 Outcome Synthesis and Interpretation

The final stage of the model translates the active scenario into directional labor-outcome tendencies. Coded evidence is used to construct qualitative rules linking combinations of institutional, organizational, and work-design conditions to greater or lower pressure on productivity, displacement pressure, skill formation, job quality, employment, redeployment capacity, and the distribution of gains.

These rules are evidence-informed rather than statistical estimates. The resulting movements do not represent effect sizes, probabilities, or forecasts, and the model does not claim that a particular condition guarantees a particular outcome. Where different mechanisms

create competing pressures, the multidimensional display allows those tensions to remain visible rather than converting them into a single overall score.

The outcomes are therefore best interpreted comparatively. Once a user reaches the outcome stage, the resulting scenario becomes Future A and can be compared with Future B, generated by changing institutional, organizational, or work-design choices while holding the selected occupation and technological baseline constant. The comparison makes visible which outcome dimensions change, which remain relatively stable, and where different configurations produce trade-offs. The purpose is not to identify a single optimal or most likely future, but to show how different choices and institutional conditions can produce divergent pathways from the same starting point.

4. COMPARATIVE CASES

To ground the framework in real institutional variation, the project considers three comparative cases: the United States, Singapore, and Denmark. These cases are not intended as one-to-one equivalents of the three institutional scenarios in the Explorer, nor are they used to claim that a particular institutional system produces a particular labor outcome. The comparison also does not assume that one system is inherently better than the others. Instead, the cases illustrate different ways of organizing responsibility for worker training, labor-market transitions, employer adjustment, and worker protection, each with its own trade-offs and institutional logic.

4.1 United States

The United States provides an example of a relatively decentralized workforce-development system. Federal programs provide funding and broad policy frameworks, but implementation is distributed across states, local workforce boards, educational institutions, training providers, employers, and other organizations. Altman and Schrag (2025) show that this produces considerable variation in how workforce development is funded and delivered. They also point to a persistent challenge for employer-sponsored training: where workers move frequently between employers, firms may be less willing to invest in skills whose benefits they may not ultimately capture. Public funding, credentials, apprenticeships, and

other workforce institutions can therefore affect whether and how workers gain access to training during periods of economic change.

For this project, the U.S. case illustrates an adjustment system in which the capacity to retrain or move workers into new roles can vary substantially across employers, programs, sectors, and places rather than being organized through a single national system.

4.2 Singapore

Singapore represents a more deliberately coordinated approach to workforce adaptation. SkillsFuture, launched in 2015, brings together government, employers, unions, training providers, industry partners, and individuals around continuing skills development. Its programs include support for individual training and career transitions as well as employer-facing initiatives for workplace learning, reskilling, and job redesign (SkillsFuture Singapore, n.d.). Singapore's broader system of tripartism also gives government, employer organizations, and unions established channels through which labor and workforce issues are addressed jointly (Ministry of Manpower, n.d.).

This coordination is now being applied directly to AI-related labor change. In 2026, the government, the National Trades Union Congress, and the Singapore National Employers Federation announced plans to form a Tripartite Jobs Council focused on AI's effects on workers, jobs, and businesses, including support for AI skills, reskilling, and job redesign (Ministry of Manpower, 2026). The significance for this project is not that coordination guarantees better outcomes, but that adaptation is organized as a shared institutional responsibility rather than being left solely to individual workers or employers.

4.3 Denmark

Denmark provides a different model of coordinated adjustment. Its labor-market system is commonly described through flexicurity, which combines relatively flexible hiring and dismissal with income support for workers and active labor-market policies intended to help people move back into employment (Danish Agency for Labour Market and Recruitment, 2026). Collective bargaining also plays an important role: many employment conditions are negotiated between unions and employer associations rather than set primarily through legislation (Ministry of Employment and Gender Equality, 2026).

This distinction matters for the framework. Denmark is not simply a case of stronger protection for existing jobs. Its institutional model allows employers considerable flexibility to reorganize their workforces while providing workers with greater support between jobs and through retraining and employment services. The OECD continues to identify active labor-market policy as a central part of Danish flexicurity, although the public employment system is currently undergoing reform (OECD, 2026b).

4.4 Comparative Implications

Taken together, the three cases show that technological change enters labor markets through very different institutional settings. The U.S. distributes much of the responsibility for adjustment across a fragmented network of public programs, employers, and local actors; Singapore builds more explicit coordination around skills, employers, unions, and government; and Denmark combines labor-market flexibility with collective bargaining, income security, and active support for worker transitions. The comparison does not establish which system will respond "best" to AI. Rather, it makes the project's central point more concrete: similar technological pressures can encounter very different systems for deciding who bears the costs of adjustment, who supports worker transitions, and how employers are able or encouraged to respond.

5. TRANSLATING THE FRAMEWORK INTO A DIGITAL ARTIFACT

One of the motivations for building the Scenario Explorer was the difficulty of making sense of the existing information about AI and work. There is no shortage of research. Academic studies, occupational datasets, policy reports, industry analyses, and media coverage all offer different ways of describing what AI may mean for workers and labor markets. Yet taken together, this information can be difficult to navigate. Measures of exposure are easily confused with predictions of job loss; economic models rely on concepts that are not always accessible outside the field; and media narratives often move quickly between enthusiasm and alarm. I spent a considerable amount of time reading across these sources simply trying to understand how the different pieces fit together.

That experience shaped the decision to build a digital artifact rather than present the project only as a written analysis. I wanted the Scenario Explorer to function as a sensemaking tool: a way of translating a complex and fragmented body of evidence into a structure that a non-specialist could explore without needing a background in labor economics, AI research, or public policy. The goal is not to simplify away uncertainty or disagreement, but to make the underlying questions and choices more legible.

This motivation also reflects the reason I entered the Data Analysis and Visualization program. I am interested in using data and visual communication not simply to analyze complex problems, but to make them understandable to people who are affected by them. Questions about AI and the future of work extend well beyond technology or economics. They concern who has opportunities, who bears the costs of technological change, how work is organized, what kinds of skills remain valuable, and how the gains from new technologies are distributed. These are questions with broad social consequences, and I do not think understanding them should require specialized expertise.

The structure of the Explorer is designed around this principle. Rather than presenting the user with a single prediction, it breaks the transition from AI capability to labor outcomes into a sequence that can be examined one layer at a time: occupational exposure, institutional environment, organizational response, human–AI work design, and labor-market outcomes. Keeping these layers separate is a communication choice as much as an analytical one. It allows users to see where different kinds of decisions enter the process instead of encountering a single number or headline suggesting that an occupation is simply "at risk."

Interactivity adds another dimension to this form of sensemaking. Users do not only read about institutional conditions or employer choices; they can change them. Occupation-specific tasks can be reorganized between Human-led, Human + AI, and AI-led work, and the resulting labor-market tendencies remain multidimensional rather than being reduced to a favorable or unfavorable score. When a user reaches Future A, they can then construct Future B by changing institutional, organizational, or work-design choices and compare the two. The comparison makes the project's central argument visible through interaction: the same technological starting point does not necessarily lead to the same future.

There is also a civic motivation behind this design. Public discussion about AI often places people in the position of spectators to a future being described by technology

companies, policymakers, economists, or media narratives. The Scenario Explorer starts from a different premise: the future of work is not predetermined, and technological capability does not settle what follows from it. Institutions can be redesigned, organizations can make different decisions, and work itself can be organized differently. Making those possibilities visible does not tell users what the future should look like, but it makes clear that alternative paths remain open.

6. DISCUSSION

One of the strongest conclusions I draw from this project is that the debate about AI and work may be asking the wrong question. Much of the discussion begins with whether AI will replace particular jobs. But that question directs attention toward technological capability while leaving less visible the decisions that shape what happens next. One thing that became hard to ignore while working on this project was the asymmetry between what appears easier to measure and what may ultimately matter. Occupational exposure can be expressed as a score and compared systematically across occupations. The cleanest numbers in the debate seem to carry the most authority, giving the implications of AI a sense of precision and inevitability. By contrast, the social and organizational processes that follow are much harder to quantify, and the concrete choices and trade-offs that influence how AI is adopted receive far less systematic study.

Building the Explorer also changed how I thought about the idea of a "better" future of work. Early versions of the project tended toward classifying outcomes as more or less favorable. That became more difficult to defend as the framework developed. The trade-offs between different outcomes are not inconsistencies to resolve; they are part of the result. This shifted the question for me from which future is "better" to what is gained, what is lost, and for whom.

These observations led me to think of the future of work less as a future to be forecast and more as a design space to be shaped. This does not deny the constraints created by technological capabilities, economic pressures, or existing institutions. It changes the question. Forecasting asks what is likely to happen. A design perspective asks what outcomes we want

to pursue, what trade-offs we are willing to accept, and who gets to participate in making those choices.

People cannot participate in shaping choices they cannot see. For me, this is the civic implication of treating the future of work as a design space. The point of making these questions more legible is not to replace expert analysis with simple answers or to tell people which future they should prefer. It is to make the decisions and competing interests visible enough to be debated. An issue that will shape working lives should not be intelligible only to specialists or narrated primarily by those with the greatest technological, economic, or institutional power. Meaningful participation begins with being able to see where choices still exist.

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